

1 Introduction

1.1 Background of the Study

Hypertension, often referred to as the "silent killer" due to its asymptomatic nature, is a leading non-communicable disease and a principal contributor to cardiovascular morbidity and premature mortality worldwide. In the Philippine context, the burden remains substantial. Recent survey cycles, specifically the 2018 National Nutrition Survey (NNS), report a prevalence of 19.2% among Filipino adults aged 20 to 59, escalating dramatically to 35.0% among the elderly. Despite a reported disease awareness rate of 67.8%, blood pressure control remains critically poor; only 75% of those aware are on treatment, and of those treated, only 27% have their blood pressure successfully controlled [1]. This gap contributes significantly to heart disease and stroke, which remain the top causes of mortality in the country.

Advances in artificial intelligence (AI) and machine learning (ML) enable scalable risk stratification and targeted public-health screening by leveraging large, multimodal datasets. These techniques have been applied across settings to predict cardiovascular risk, specifically utilizing the National Nutrition Survey (NNS) provided by the DOST-FNRI [2], and successful deployments as accessible web applications have been demonstrated [3, 4].

Two practical concerns limit the clinical utility of many ML models: (1) model explainability, and (2) probability reliability. Explainable AI (XAI) methods such as SHAP and LIME increase transparency and clinician trust [5, 6]. Independently, probability calibration techniques ensure that a model's predicted probabilities reflect observed frequencies, a requirement when predictions are used in decision-making workflows. Recent methodological work integrates these concerns through the Calibrated Explanations framework, which produces uncertainty-aware explanations and Wachter-style counterfactuals optimized for minimal clinical perturbation aligned with calibrated probabilities [7, 8].

Turnitin Draft Coach

Details Similarity Citations Grammar

11%
Overall Similarity
Last checked: May 20 1:39 PM PST

What should I do with my score?

1	www.mdpi.com	1%
INTERNET		
2	journalisra.com	1%
INTERNET		
3	www.researchsquare.c...	<1%
INTERNET		

You can run 2 more similarity checks.

Run New Similarity Check

View Full Report

How should I use these checks?